Appendix

Anonymous ECCV 2026 Submission

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A Simulate the performance on an unseen set

Original Model Training

We train the model on the CIFAR-100 dataset, using a **ViT-B/32** vision backbone optimized with the **AdamW** optimizer. The batch size is set to 64, and the model is trained for 50 epochs. The optimizer uses momentum parameters $\beta_1 = 0.9$ and $\beta_2 = 0.98$, with a numerical stability term $\epsilon = 1 \times 10^{-6}$. The initial learning rate is 5×10^{-6}, and a **cosine learning rate scheduler** is applied. We employ a warmup phase of 10,000 steps, followed by cosine decay with `lr_cooldown_end` $= 0.0$ and `lr_cooldown_power` $= 1.0$. Weight decay is set to 0.1 for regularization. Mixed-precision training (**AMP**) is adopted to improve computational efficiency. The temperature parameter is set to $\tau = 0.07$. 30% of the dataset is reserved as the validation split.

Noise Simulation

Since the true unseen distribution is inaccessible in practice, we approximate it using isotropic noise inputs that are statistically independent of the training semantics. Under the assumption that noise embeddings are approximately uniformly distributed on the hypersphere, their cosine similarity with learned semantic directions should concentrate around zero. Any systematic deviation indicates residual structural bias in the representation space, revealing incomplete forgetting or latent semantic alignment. To verify this hypothesis, we conduct the following two experiments.

1. **Random Text → Real Images:** This experiment evaluates how real images respond to non-semantic textual inputs. We evaluate the model on the test set of CIFAR-100. For each class, we randomly sample 100 test images and generate 20 random character sequences (random length in range 10) as non-semantic text inputs. Image and text embeddings are extracted using the trained model and ℓ_2-normalized. Cosine similarity is computed as:

$$\text{sim}(x_i, t_j) = f(x_i)^\top f(t_j) \tag{1}$$

   where $f(\cdot)$ denotes the normalized embedding function.
2. **Random Images → Real Text:** This experiment evaluates how semantic text directions respond to non-semantic visual inputs.We generate synthetic noise images Gaussian noise and uniform noise. We generate 30 noise

images and compute their cosine similarities with 100 textual labels using the prompt template "a picture of label."Image and text embeddings are extracted using the trained model and ℓ_2-normalized. Cosine similarity is computed as Eq. (1)

B Baseline Experiments

Gradient Ascent [8]

The gradient ascent method modifies model parameters to make the output for specific inputs deviate from the expected result, effectively causing the model to forget this knowledge. Unlike gradient descent, which minimizes the loss, gradient ascent maximizes it during training. Formally, let x be a sample from the forget set D_F, and $\ell(x, \theta)$ denote the loss for this sample given model parameters θ. The model parameters are adjusted by maximizing the loss function $\mathcal{L}_{\text{GA}}$:

$$\mathcal{L}_{\text{GA}}(D_F, \theta) = \frac{1}{|D_F|} \sum_{x \in D_F} \ell(x, \theta)$$

By ascending the gradient of the loss on the forget set, the model is pushed away from its original predictions, thereby unlearning the target knowledge.

Gradient Difference [4]

This method combines gradient ascent on the forget set with gradient descent on the retain set to preserve performance on the retain set. The model parameters are updated by minimizing the following joint loss function:

$$\mathcal{L}_{\text{GD}}(D_F, D_R, \theta) = -\frac{1}{|D_F|} \sum_{x \in D_F} \ell(x, \theta) + \frac{1}{|D_R|} \sum_{x \in D_R} \ell(x, \theta)$$

By ascending the gradient on the forget set and descending on the retain set, the model forgets the target knowledge while maintaining utility on the retained data.

KL Minimization [7]

The KL minimization method calculates the Kullback-Leibler (KL) divergence between the original and unlearned models' predictions on the retain set, while maximizing the loss on forget set. Let $\mathcal{M}$ denote the model and $\mathcal{M}(\cdot)$ its output probability distribution. The loss function $\mathcal{L}_{\text{KL}}$ is formulated as:

$$\mathcal{L}_{\text{KL}} = -\mathcal{L}(D_F, \theta) + \frac{1}{|D_R|} \sum_{s \in D_R} \frac{1}{|s|} \sum_{i=2}^{|s|} \text{KL}\left(\mathcal{M}_\theta(s_{<i}) \parallel \mathcal{M}_{\hat{\theta}}(s_{<i})\right) \tag{7}$$

Here, the first term $-\mathcal{L}(D_F, \theta)$ performs gradient ascent on the forget set D_F to suppress unwanted knowledge, while the second term minimizes the KL divergence between the original model $\mathcal{M}_\theta$ and the unlearned model $\mathcal{M}_{\hat{\theta}}$ on the retain set D_R. The KL divergence is computed over each token position i in sequence s, conditioning on the previous tokens $s_{<i}$. This dual-objective formulation balances forgetting effectiveness with utility preservation, ensuring that the unlearned model maintains performance on retained data while forgetting target information.

C Experiment Settings

Dataset and Hyperparameters

Our implementation is built upon the open-source OpenCLIP [2] framework. We conducted our experiments on three Datasets, **CIFAR-100** [3], **Flickr30k** [9] and **CC12M** [1]. All models are trained on an NVIDIA V100 GPU using a ViT-B/32 vision backbone and optimized with AdamW. The optimizer uses momentum parameters $\beta_1 = 0.9$ and $\beta_2 = 0.98$. Random seed is set as 42. We first fine-tune the model on the dataset using the standard CLIP loss function to obtain a trained baseline model for comparison. Then, we apply different forgetting methods to the weights of this baseline model to selectively erase specific concepts.

1. **CIFAR-100:** The dataset consists of 60,000 images, encompassing 100 categories, with 600 images per category. These images are divided into a training set and a test set. The training set contains 50,000 images across all 100 categories, while the test set comprises 10,000 images from the same 100 categories. The batch size is set to 64, and the model is trained for 10 epochs. The initial learning rate is 1×10^{-6}, and a cosine learning rate scheduler is applied. The loss balancing coefficients are set as $\alpha_1 = 0.5$, $\alpha_2 = 1.0$, $\alpha_3 = 1.0$, and $\alpha_4 = 3.0$.
2. **Flickr30k:** The dataset contains 31,783 images collected from the Flickr website. Each image is paired with 5 independent descriptive sentences. We divide it into training, validation, and test sets in a 7:2:1 ratio. The batch size is set to 64, and the model is trained for 20 epochs. The initial learning rate is 1×10^{-7}, and a cosine learning rate scheduler is applied. The loss balancing coefficients are set as $\alpha_1 = 2.0$, $\alpha_2 = 0.5$, $\alpha_3 = 1.0$, and $\alpha_4 = 3.0$.
3. **CC12M:** The dataset contains approximately 2 million image-text pairs collected from web sources. We use it to simulate real-world pre-training dataset containing noise. In this experiment, around 10% of the dataset is held out as the material for unlearning. The batch size is set to 128, and the model is trained for 20 epochs. The initial learning rate is 1×10^{-6}, and a cosine learning rate scheduler is applied. The loss balancing coefficients are set as $\alpha_1 = 0.5$, $\alpha_2 = 1.0$, $\alpha_3 = 1.0$, and $\alpha_4 = 3.0$.

Task and Testset

For model evaluation, we mainly use two datasets: **CIFAR-100** and **Flickr30k**. For each training run, we select the model with the lowest validation loss as the optimal model.

Evaluation on CIFAR-100 : Following the standard protocol, we use the official test split containing 10,000 images for evaluation. To assess the forgetting and retention capabilities of our model, we designate a subset of classes as the forget set, while the remaining classes form the retain set. We conduct a comprehensive evaluation from multiple perspectives: zero-shot classification accuracy, measuring how well the model predicts the correct class labels; zero-shot retrieval performance, evaluated by Recall@1, @5, and @10 metrics to assess cross-modal alignment; cosine similarity distribution analysis, which examines the feature space separation between forget and retain samples; attack simulation, using minimum cosine similarity for image-text matching in simulated real-world attack scenarios.

Evaluation on Flickr30k : We evaluate on the Flickr30K dataset with a targeted forgetting scenario centered on the concept "*mountain*". The forget set $\mathcal{D}_f$ consists of 500 mountain-related images with their captions, while the retain set $\mathcal{D}_r$ comprises the remaining images. We conduct three evaluations:

- **Image Retrieval (Text→Image)**: Using mountain-specific text queries (Tab. 2 and Tab. 3) and random retain queries, we measure Recall@k for retrieving forget/retain images.
- **Text Retrieval (Image→Text)**: Using mountain and non-mountain images as queries, we measure Recall@k for retrieving correct captions.
- **Attack Success Rate**: Using *minimum similarity search*, we measure how easily adversarial queries can discover forget-set samples.

Image Generation task on Stable Diffusion [5]: We employ Stable Diffusion v1.5 as our base generative model. The pipeline is configured to run on CUDA with float16 precision for optimal efficiency, with the safety checker disabled as it is not required for our experimental purposes. To evaluate the effect of unlearning, we replace the original CLIP text encoder with a custom fine-tuned version. This custom encoder is loaded using Hugging Face's CLIPTextModel and CLIPTokenizer interfaces, ensuring compatibility with the diffusion pipeline architecture. The replacement process preserves all other components of the pipeline unchanged, allowing us to isolate the impact of text encoder modifications on image generation quality.

To align with the model weights of Stable Diffusion, we select the ViT-L/14 CLIP architecture as the backbone for unlearning. The model is fine-tuned on the CIFAR dataset with a specific focus on forgetting the concept "apple". The batch size is set to 64, and the model is trained for 20 epochs. The initial

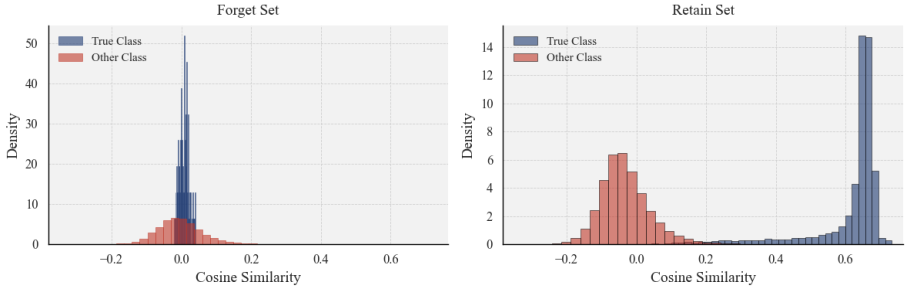


Fig. 1: TCDE unlearned model:cosine similarity distributions,between *Positive pair (True)* (image paired with text from same category) and *Negative pair (Other)* (image paired with text from different categories).

learning rate is 1×10^{-6}, and a cosine learning rate scheduler is applied. The loss balancing coefficients are set as $\alpha_1 = 1.0$, $\alpha_2 = 1.0$, $\alpha_3 = 1.0$, and $\alpha_4 = 3.0$.

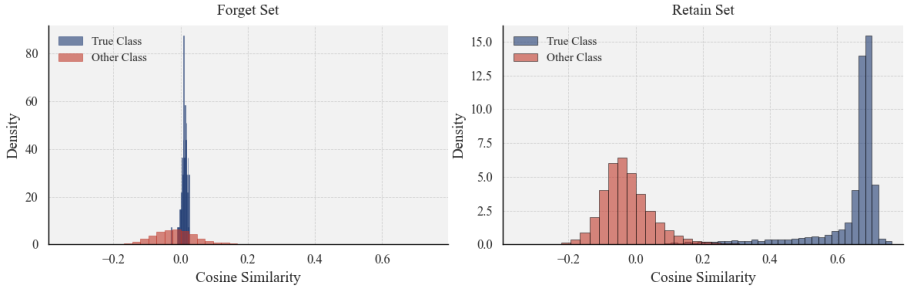
For consistent and reproducible results, we configure the generation process with 40 inference steps and a guidance scale of 7.5, which balances generation quality with prompt adherence. A fixed random seed of 42 is used to ensure reproducibility across experiments.

Comparison of cosine similarity statistics: Fig. 3 and Fig. 1 below present a statistical comparison of cosine similarity distributions on the forget and retain sets, obtained after applying various baseline unlearning methods GA, GD, KL, CLIPErase and our method TCDE to erase the concept "apple" on the CIFAR-100 dataset.

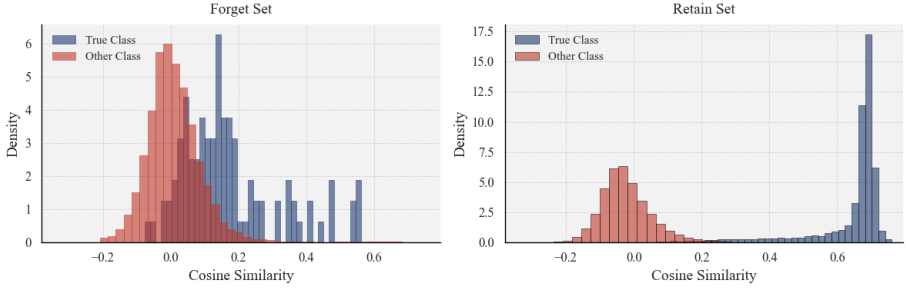
Based on the cosine similarity statistics of each method, we observe that after unlearning, the similarity scores of almost all forget-set image-text pairs processed by the baseline methods are concentrated in the range of -1 to -0.8. Meanwhile, a small portion of pairs that fail to be unlearned retain high similarity, indicating correct alignment. These results reveal that forgetting is either traceable or incomplete. Therefore, as illustrated in the cosine similarity distribution, when the baseline unlearned model is applied to classification or retrieval tasks, an attacker can readily discern whether the model has undergone unlearning and can precisely recover the forgotten targets by selecting those with the minimum cosine similarity.

D Ablation experiments

We conduct ablation studies to evaluate the contribution of each loss component in our proposed TCDE framework. Specifically, we investigate the effectiveness of the Energy Distance Loss ($\mathcal{L}_{\text{DIST}}$), Hypersphere Uniformity Loss ($\mathcal{L}_{\text{UNIF}}$), Retention Loss ($\mathcal{L}_{\text{RET}}$), and Consistency Loss ($\mathcal{L}_{\text{CM}}$) by systematically removing each component from the full objective function. This is achieved by setting the



(a) Without Hypersphere Uniformity Loss ($\alpha_2 = 0$)



(b) Without Energy Distance Loss ($\alpha_1 = 0$)

Fig. 2: Comparison of cosine similarity distributions on positive and negative pairs after applying unlearn methods to erase the concept "apple": (a) model trained without Hypersphere Uniformity Loss; (b) model trained without Energy Distance Loss.

corresponding hyperparameter $(\alpha_1, \alpha_2, \alpha_3, \alpha_4)$ to zero while keeping the others at their optimal values. By comparing the performance of these ablated variants against the full model, we assess the individual impact of each loss on both unlearning effectiveness and traceability mitigation. In our experimental setup, we conduct concept unlearning experiments on the CIFAR-100 dataset, specifically targeting the concept of "apple" for erasure. All experimental parameters, except for the hyperparameters under investigation, are kept consistent across all ablation variants. The batch size is set to 64, and the model is trained for 10 epochs. The initial learning rate is 1×10^{-6}, and a cosine learning rate scheduler is applied.

We conduct ablation studies to analyze the contribution of each loss component in our proposed TCDE framework, with results shown in Tab. 1. Several key observations can be drawn from these experiments.

First, the Retention Loss ($\mathcal{L}_{\text{RET}}$) and Consistency Loss ($\mathcal{L}_{\text{CM}}$) primarily contribute to the stability of performance on the retain set. When $\mathcal{L}_{\text{RET}}$ is removed ($\alpha_3 = 0$), we observe a slight degradation in retain set accuracy (Acc.Dr drops from 88.34% to 87.77%) and retrieval performance (R@5 decreases from 98.20% to 97.93%). Similarly, removing $\mathcal{L}_{\text{CM}}$ ($\alpha_4 = 0$) leads to minor fluctuations in retain set metrics. This stabilizing effect becomes more pronounced in multi-class

Table 1: Ablation studies on CIFAR-100 for single class forgetting (apple). Each variant removes one loss component by setting its corresponding hyperparameter to zero. Acc.Df/Acc.Dr: Accuracy on forget/retain classes; R@k: Recall@k, image to text retrieval; Desired direction: ↓ lower is better, ↑ higher is better.

Method	Acc.Df↓	Acc.Dr↑	Forget Set		Retain Set		ASR ↓
			R@5↓	R@10↓	R@5↑	R@10↑	(%)
Single Class Forgetting (Apple)							
Original	94.0	87.78	100.0	100.0	97.97	98.78	-
Ablation Variants ($\alpha_i = 0$)							
x $\mathcal{L}_{\text{DIST}}$ ($\alpha_1 = 0$)	23.0	88.28	49.0	64.0	98.19	99.02	0.0
x $\mathcal{L}_{\text{UNIF}}$ ($\alpha_2 = 0$)	0.0	88.42	0.0	0.0	98.22	99.07	0.0
x $\mathcal{L}_{\text{RET}}$ ($\alpha_3 = 0$)	0.0	87.77	0.0	0.0	97.93	98.77	0.0
x $\mathcal{L}_{\text{CM}}$ ($\alpha_4 = 0$)	0.0	88.19	0.0	0.0	98.14	99.03	0.0
TCDE (Full)	**0.0**	**88.34**	**0.0**	**0.0**	**98.20**	**99.07**	**0.0**

forgetting scenarios, where the retention of knowledge across multiple concepts becomes increasingly challenging.

Second, the Energy Distance Loss ($\mathcal{L}_{\text{DIST}}$) plays a critical role in ensuring thorough forgetting and proper alignment of cosine similarity distributions. When $\mathcal{L}_{\text{DIST}}$ is ablated ($\alpha_1 = 0$), we observe a significant degradation in forgetting effectiveness: Acc.Df increases to 23.0%, and forget set retrieval metrics rise dramatically (R@5 reaches 49.0%, R@10 reaches 64.0%). This confirms that $\mathcal{L}_{\text{DIST}}$ is essential for making forget-set samples statistically indistinguishable from the retain set. Notably, unlike conventional unlearning methods that forcibly push forget-set pairs arbitrarily far apart, our approach with $\mathcal{L}_{\text{DIST}}$ avoids the extreme concentration of cosine similarities near -1, thereby maintaining the Attack Success Rate (ASR) at 0.0% across all ablation variants.

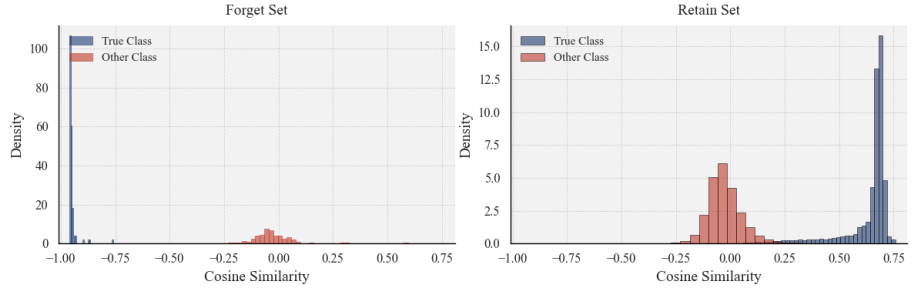
Third, the Hypersphere Uniformity Loss ($\mathcal{L}_{\text{UNIF}}$) demonstrates a subtle but important role in shaping the distribution of embeddings. Interestingly, when $\mathcal{L}_{\text{UNIF}}$ is removed ($\alpha_2 = 0$), the retain set performance appears marginally better (Acc.Dr of 88.42% vs. 88.34% in the full model). However, examination of the cosine similarity distribution plots (see Fig. 1 and Fig. 2a) reveals a critical issue: without $\mathcal{L}_{\text{UNIF}}$, the forget set exhibits an abnormal peak near zero cosine similarity for positive pairs. This conspicuous artifact in the similarity distribution could potentially be exploited for detecting unlearning activity, compromising the untraceability objective. Therefore, while $\mathcal{L}_{\text{UNIF}}$ may not significantly impact aggregate metrics, it plays an indispensable role in eliminating detectable traces and ensuring that forget-set samples blend seamlessly with the statistical patterns of unseen concepts.

In summary, each loss component contributes uniquely to the overall performance: $\mathcal{L}_{\text{DIST}}$ ensures thorough forgetting, $\mathcal{L}_{\text{UNIF}}$ maintains distributional indistinguishability, while $\mathcal{L}_{\text{RET}}$ and $\mathcal{L}_{\text{CM}}$ stabilize retain set performance. The

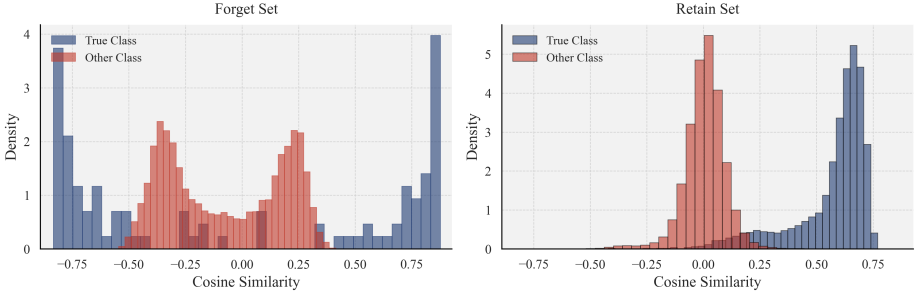
full TCDE model achieves the best balance across all objectives, with zero forget set accuracy and perfect traceability mitigation.

Table 2: Mountain-specific text queries used for forget set evaluation on Flickr30K. (Part 1/2)

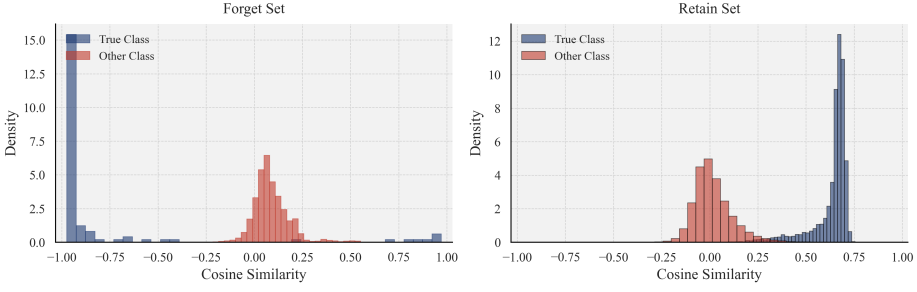
Mountain-Related Text Queries	
a picture of mountains	a photo of mountain peaks
a view of hills and mountains	mountain landscape
snowy mountains	rocky mountains
mountain scenery	mountains and sky
mountain range	alpine mountains
high mountains	green mountains
mountain view	mountainous terrain
mountain slope	a majestic mountain with snow capped peaks
rugged mountain terrain with cliffs	volcanic mountain with smoke
mountain pass between two peaks	mountain ridge at sunset
steep mountain face with rock formations	rolling mountains in the distance
mountain valley with river	mountain summit covered in clouds
mountain foothills with trees	mountain plateau with grass
mountain canyon with steep walls	mountain glacier and ice fields
mountain lake surrounded by peaks	mountain meadow with wildflowers
mountain desert landscape	mountain tundra with sparse vegetation
mountain forest on the slopes	mountain waterfall cascading down rocks
mountain hot springs	sunny day in the mountains
cloudy mountains with mist	rainy mountain landscape
foggy mountains in the morning	snowstorm in the mountains
mountains during golden hour	mountain sunset with orange sky
mountain sunrise over peaks	moonlit mountain landscape
mountains in autumn with colorful trees	spring mountains with melting snow
summer mountains with green vegetation	winter mountains completely covered in snow
mountains after rain with clear air	mountains with rainbow after storm
extreme high altitude mountains	frozen mountain landscape
barren mountain desert	tropical mountain rainforest
coastal mountains meeting ocean	sacred mountain in local culture
legendary mountain from myths	historical mountain fortress
mountain pilgrimage site	mountain with spiritual significance



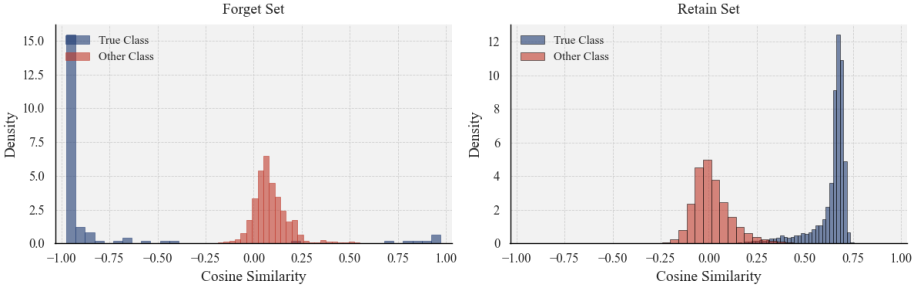
(a) CLIPErase unlearned model



(b) GA unlearned model



(c) GD unlearned model



(d) KL unlearned model

Fig. 3: Comparison of cosine similarity distributions on positive and negative pairs after applying unlearn methods to erase the concept "apple": (a) CLIPErase [6]; (b) GA; (c) GD; (d) KL.

Table 3: Mountain-specific text queries used for forget set evaluation on Flickr30K. (Part 2/2)

Mountain-Related Text Queries	
the Himalayan mountains	the Rocky Mountains
the Alps mountain range	the Andes mountains
the Appalachian Mountains	the Sierra Nevada mountains
the Ural Mountains	the Caucasus Mountains
the Carpathian Mountains	the Pyrenees mountains
the Atlas Mountains	the Tian Shan mountains
the Kunlun Mountains	the Drakensberg mountains
the Scottish Highlands	aerial view of mountains
close up of mountain rocks	panoramic mountain vista
mountains reflected in a lake	mountains from a valley below
mountains seen through trees	mountains with a small cabin
mountains with hiking trail	mountains with cable car
mountains with winding road	mountains with ancient ruins
mountains with monastery on cliff	mountains with village at base
mountains with wildlife	mountains with climbers
painting of mountains	black and white mountain photography
abstract mountain shapes	impressionist mountain scene
surreal mountain landscape	minimalist mountain composition
mountain silhouette against sky	textured mountain surface
geometric mountain patterns	dreamlike mountain imagery
peaceful mountain retreat	majestic and grand mountains
lonely mountain in wilderness	forbidding mountain landscape
serene mountain scenery	powerful mountain presence
ancient and timeless mountains	mysterious mountain fog
tranquil mountain morning	dramatic mountain storm
mountain climbing expedition	skiing down mountain slopes
hiking in mountain trails	camping in mountain wilderness
mountain biking on dirt trails	paragliding over mountains
mountain railway journey	fishing in mountain stream
photographing mountain wildlife	mountain rescue operation

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